

ENVIRONMENTAL SCIENCE



Environmental Science, Grade 11

Workplace Preparation

SVN3E

This course provides students with the fundamental knowledge of and skills relating to environmental science that will help them succeed in work and life after secondary school. Students will explore a range of topics, including the impact of human activities on the environment; human health and the environment; energy conservation; resource science and management; and safety and environmental responsibility in the workplace. Emphasis is placed on relevant, practical applications and current topics in environmental science, with attention to the refinement of students' literacy and mathematical literacy skills as well as the development of their scientific and environmental literacy.

Prerequisite: Science, Grade 9, Academic or Applied, or a Grade 9 or 10 locally developed compulsory credit (LDCC) course in science

Big Ideas

Human Impact on the Environment

- The biotic elements of an ecosystem include humans.
- Abiotic and biotic factors interact within an ecosystem.
- People have positive and negative effects on the environment, both locally and globally.

Human Health and the Environment

- Environmental factors can have negative effects on human health.
- It is possible to minimize some negative health effects associated with environmental factors by making wise lifestyle choices and taking other precautions.

Energy Conservation

- The impact of energy production and consumption on environmental sustainability depends on which resources and energy production methods are used.

Natural Resource Science and Management

- Biodiversity is a measure of the health and sustainability of an ecosystem.
- Careful resource management planning is necessary to sustain ecosystems.

The Safe and Environmentally Responsible Workplace

- Workplace safety is the responsibility of both employees and employers.

Fundamental Concepts Covered in This Course (see also page 5)

Fundamental Concepts	Human Impact on the Environment	Human Health and the Environment	Energy Conservation	Natural Resource Science and Management	The Safe and Environmentally Responsible Workplace
Matter	✓	✓		✓	✓
Energy		✓	✓		
Systems and Interactions	✓				✓
Structure and Function					
Sustainability and Stewardship	✓	✓	✓	✓	✓
Change and Continuity	✓			✓	

A. SCIENTIFIC INVESTIGATION SKILLS AND CAREER EXPLORATION

OVERALL EXPECTATIONS

Throughout this course, students will:

- A1.** demonstrate scientific investigation skills (related to both inquiry and research) in the four areas of skills (initiating and planning, performing and recording, analysing and interpreting, and communicating);
- A2.** identify and describe careers related to the fields of science under study, and describe contributions of scientists, including Canadians, to those fields.

SPECIFIC EXPECTATIONS

A1. Scientific Investigation Skills

Throughout this course, students will:

Initiating and Planning [IP]*

- A1.1** formulate relevant scientific questions about observed relationships, ideas, problems, or issues, make informed predictions, and/or formulate educated hypotheses to focus inquiries or research
- A1.2** select appropriate instruments (e.g., pH probes, plant tags, soil markers) and materials (e.g., botanical keys, personal protection devices, soil test kits), and identify appropriate methods, techniques, and procedures, for each inquiry
- A1.3** identify and locate a variety of print and electronic sources (e.g., Material Safety Data Sheets, appliance manuals, hydro bills, the Live Safe! Work Smart! website) that enable them to address research topics fully and appropriately
- A1.4** apply knowledge and understanding of safe laboratory practices and procedures when planning investigations by correctly interpreting Workplace Hazardous Materials Information System (WHMIS) symbols; by using appropriate techniques for handling and storing laboratory equipment and materials and disposing of laboratory materials; and by using appropriate personal protection

Performing and Recording [PR]*

- A1.5** conduct inquiries, controlling relevant variables, adapting or extending procedures as required, and using appropriate materials and equipment safely, accurately, and effectively, to collect observations and data
- A1.6** compile accurate data from laboratory and other sources, and organize and record the data, using appropriate formats, including tables, flow charts, graphs, and/or diagrams
- A1.7** select, organize, and record relevant information on research topics from a variety of appropriate sources, including electronic, print, and/or human sources, using suitable formats and an accepted form of academic documentation

Analysing and Interpreting [AI]*

- A1.8** synthesize, analyse, interpret, and evaluate qualitative and/or quantitative data to determine whether the evidence supports or refutes the initial prediction or hypothesis and whether it is consistent with scientific theory; identify sources of bias and/or error; and suggest improvements to the inquiry to reduce the likelihood of error
- A1.9** analyse the information gathered from research sources for logic, accuracy, reliability, adequacy, and bias
- A1.10** draw conclusions based on inquiry results and research findings, and justify their conclusions with reference to scientific knowledge

* The abbreviation(s) for the broad area(s) of investigation skills – IP, PR, AI, and/or C – are provided in square brackets at the end of the expectations in strands B–F to which the particular area(s) relate (see pp. 20–22 for information on scientific investigation skills).

Communicating [C]*

A1.11 communicate ideas, plans, procedures, results, and conclusions orally, in writing, and/or in electronic presentations, using appropriate language and a variety of formats (e.g., data tables, laboratory reports, presentations, graphic organizers, simulations, models, workplace labels)

A1.12 use appropriate numeric, symbolic, and graphic modes of representation, and appropriate units of measurement (e.g., SI and imperial units)

A1.13 express the results of any calculations involving data accurately and precisely, to the appropriate number of decimal places or significant figures

A2. Career Exploration

Throughout this course, students will:

A2.1 identify and describe a variety of careers related to the fields of science under study (e.g., hydro meter reader, hospitality employee, waste management operator, custodian) and the education and training necessary for these careers

A2.2 describe the contributions of scientists, including Canadians (e.g., Mark Schacter, Sheila Watt-Cloutier, Marlo Reynolds, J. Ross MacKay, Linda Duncan), to the fields under study

B. HUMAN IMPACT ON THE ENVIRONMENT

OVERALL EXPECTATIONS

By the end of this course, students will:

- B1.** analyse selected current environmental problems in terms of the role human activities have played in creating or perpetuating them, and propose possible solutions to one such problem;
- B2.** investigate air, soil, and water quality in natural and disturbed environments, using appropriate technology;
- B3.** demonstrate an understanding of some of the ways in which human activities affect the environment and how the impact of those activities is measured and monitored.

SPECIFIC EXPECTATIONS

B1. Relating Science to Technology, Society, and the Environment

By the end of this course, students will:

- B1.1** propose possible solutions, on the basis of research, to a current practical environmental problem that is caused, directly or indirectly, by human activities [IP, PR, AI, C]

Sample issue: Car emissions contribute to smog as well as global warming. Road tolls and increased use of public transit to cut down on the number of cars on the road, and the implementation and enforcement of idling by-laws, could significantly cut these emissions.

Sample questions: How can various kinds of chemical spills in local ecosystems (e.g., fields, rivers, streams) be cleaned up? In what ways does improper sewage treatment or agricultural run-off threaten local water supplies, and how can these dangers be addressed or averted? What can be done to minimize the effect of an invasive species (e.g., purple loosestrife) on a native species (e.g., milkweed)?

- B1.2** analyse the risks and benefits to the environment of human recreational activities and the leisure industry [AI, C]

Sample issue: Ecotourism attempts to reduce the waste and environmental damage associated with mass tourism. Although responsible ecotourism seeks to conserve local ecosystems through sustainable practices, and can, for

example, help reduce deforestation and animal poaching rates, any human intrusion can damage fragile ecosystems.

Sample questions: What are the risks to the environment of herbicide use and water consumption on golf courses? What are some of the risks and benefits to the environment of landscaping? In what ways can hunters and fishers damage the environment? In what ways can they contribute to its sustainability? What rules are needed to ensure that visitors to a protected area do not harm that ecosystem?

B2. Developing Skills of Investigation and Communication

By the end of this course, students will:

- B2.1** use appropriate terminology relating to the environmental impact of human activity, including, but not limited to: *carbon footprint*, *carbon neutral*, *biodegradable*, *biodiversity*, *carrying capacity*, *sustainability*, and *invasive and native species* [C]
- B2.2** plan and conduct an inquiry, using appropriate technology, to compare soil quality in natural and disturbed environments (e.g., compare the phosphorous content, pH, organic matter content, water content, water-holding capacity, nutrient content, porosity, and/or bulk density of soil from a forest or meadow and soil from a garden or farmer's field that has been treated with chemical fertilizer) [IP, PR, AI]

B2.3 plan and conduct an inquiry, using appropriate technology, to compare water quality in natural and disturbed environments (e.g., compare the pH, ion content, temperature, dissolved oxygen content, hardness, turbidity, biological oxygen demand [BOD], and/or fecal coliform of tap water, water from a pond or stream, and water from a drainage ditch) [IP, PR, AI]

B2.4 analyse and interpret data on particulate matter in air samples from several different regions of Canada, using prepared data from a variety of sources (e.g., the Ontario Ministry of the Environment – Air Quality Ontario, Environment Canada) [AI]

B2.5 plan and conduct a waste audit of their home or school [IP, PR]

B3. Understanding Basic Concepts

By the end of this course, students will:

B3.1 identify the basic components of soil, water, and air, and describe some of the effects of human activity on soil, water, and air quality (e.g., the effects of industrial or vehicle emissions on air quality; of chemical spills on soil quality; of chlorination on water quality)

B3.2 explain the concept of the cycling of substances in ecosystems (e.g., fertilizers made from biosolids leach into ground water or run off into rivers and streams, where the chemicals are absorbed by aquatic life, which is in turn consumed by humans)

B3.3 explain common methods of sampling soil, water, and air for analysis (e.g., soil core sampling, depth integrated sampling, stack sampling systems) and of monitoring soil, water, and air quality over time

B3.4 explain the concept of a “carbon footprint” and how it is used to measure the impact on the environment of a range of human activities

B3.5 explain the effects of human activity on an aquatic or terrestrial ecosystem (e.g., the impact of fertilizer run-off, acid precipitation, or an oil spill on an aquatic ecosystem)

B3.6 explain how human activities (e.g., agriculture, travel, the purchase of exotic pets, importing and exporting, releasing domesticated fish into fresh water environments, the use of live bait) have led to the introduction of invasive species, and why it is important to measure and monitor the impact of invasive species on native species

C. HUMAN HEALTH AND THE ENVIRONMENT

OVERALL EXPECTATIONS

By the end of this course, students will:

- C1.** analyse the effects on human health of environmental contaminants and a significant environmental phenomenon;
- C2.** investigate how different environmental factors can affect people's health and their lifestyle choices;
- C3.** demonstrate an understanding of the ways in which environmental factors can affect human health and how their impact can be reduced.

SPECIFIC EXPECTATIONS

C1. Relating Science to Technology, Society, and the Environment

By the end of this course, students will:

- C1.1** assess, on the basis of research, the effects on human health of a significant environmental phenomenon (e.g., the ice storm of 1998 in central Canada, the European heatwave of 2003), and communicate their findings [IP, PR, AI, C]

Sample issue: In August 2005, Hurricane Katrina destroyed the levees surrounding New Orleans. Hundreds of people were drowned in the resulting floods. Many survivors contracted skin and gastro-intestinal diseases from contaminated water.

Sample questions: What impact did the 2003 drought in the Okanagan Valley have on the health and well-being of local populations? How did the 2003 tsunami in the Indian Ocean affect the health of people in that region?

- C1.2** analyse how environmental contaminants can affect the health of different populations in Canada (e.g., mercury contamination in streams and rivers in Northern Ontario where Aboriginal people fish, toxins in Arctic sea mammals hunted by Inuit, smog in large cities) [AI, C]

Sample issue: When the U.S. government abandoned its Cold War military bases in the Canadian North, it left behind a variety of contaminants, including large amounts of polychlorinated biphenyls (PCBs). Exposure to these chemicals can affect the nervous system and the immune system and can cause cancer.

Sample questions: How does the use of biosolids as fertilizer on Canadian farms affect the health of local populations? What short- and long-term health problems can be traced to the chemicals in the tar ponds in Sydney, Nova Scotia?

C2. Developing Skills of Investigation and Communication

By the end of this course, students will:

- C2.1** use appropriate vocabulary related to human health and the environment, including, but not limited to: *smog, environmental contaminants, pathogens, inhalation, ingestion, and absorption* [C]
- C2.2** investigate, using a research process, and report on an environmental factor that can have an impact on human health (e.g., smog, ultraviolet [UV] rays, bacteria, pesticide residue), and explain how their personal lifestyle choices can affect its impact (e.g., avoiding strenuous physical activity on days when there is a smog alert can reduce the severity of respiratory ailments; lying on the beach without sunscreen or sun protective clothing during peak UV hours can increase the risk of skin cancer) [IP, PR, AI, C]
- C2.3** investigate the characteristics of a personal protective device or substance (e.g., sunscreen, mosquito repellent, respiratory mask, sun protective clothing) and whether the device or substance is effective in protecting a person from an environmental factor that can affect human health [PR, AI]

C3. Understanding Basic Concepts

By the end of this course, students will:

C3.1 describe common environmental factors, including pollution and environmental contaminants (e.g., air, noise, soil, and water pollution; UV rays; heat; heavy metals; workplace chemicals; pathogens), and explain how they can affect human health

C3.2 describe various ways in which environmental contaminants can enter the human body (e.g., inhalation, ingestion, absorption)

C3.3 explain how the human body can react to exposure to a variety of environmental factors

(e.g., rashes, asthma, mercury poisoning, hearing loss, diseases such as malaria and cancer)

C3.4 describe medical and non-medical ways to protect oneself from the effects of harmful environmental factors (e.g., vaccination or medication, washing of fruits and vegetables, use of sunscreen or insect repellent, use of personal protective devices)

C3.5 describe good personal hygiene and household cleanliness practices that reduce health risks resulting from environmental contaminants (e.g., thorough hand washing, use of air filters, reduced use of household chemicals)

D. ENERGY CONSERVATION

OVERALL EXPECTATIONS

By the end of this course, students will:

- D1.** evaluate initiatives and technological innovations related to energy consumption and conservation, and assess their impact on personal lifestyles, social attitudes, and the environment;
- D2.** investigate various methods of conserving energy and improving energy efficiency;
- D3.** demonstrate an understanding of the basic principles of energy production, with reference to both renewable and non-renewable sources, and of various methods of energy conservation.

SPECIFIC EXPECTATIONS

D1. Relating Science to Technology, Society, and the Environment

By the end of this course, students will:

- D1.1** assess, on the basis of research, the impact that initiatives for reducing energy consumption and waste have on personal lifestyles, societal attitudes, and the environment (e.g., local, provincial, or national initiatives by government, business, or non-governmental organizations) [IP, PR, AI, C]

Sample issue: Home energy audit and retrofit rebate programs have been established by many provincial governments to help homeowners reduce their energy bills. Although these programs raise awareness of the environmental impact of wasting energy and provide practical ways of reducing waste, not all homeowners take advantage of them.

Sample questions: What types of incentives exist to encourage consumers to purchase energy-efficient products and services? How effective are such incentives? What methods do energy companies use to encourage consumers to conserve energy? What are some of the non-governmental organizations in Canada that raise awareness of the environmental costs of energy consumption? Are there any groups in your local community that focus on energy conservation? How effective are they?

- D1.2** evaluate, on the basis of research, some of the advantages or disadvantages of technological innovations that contribute to the production of renewable energy and/or aid in conservation (e.g., bio-oil, biodiesel, wind turbines, improved insulation, programmable thermostats) [IP, PR, AI, C]

Sample issue: Tankless water heaters heat water only when it is needed. They save energy over traditional water heaters, which keep a large tank of water hot at all times. However, tankless water heaters may not be able to supply enough hot water for multiple uses.

Sample questions: What technologies are used to produce biofuels? How do these fuels help to reduce use of non-renewable energy? What problems might be associated with the use of agricultural crops for fuel rather than food? In what ways has the design of wind farm technology improved over the years? What are the advantages and disadvantages of replacing old appliances with more energy-efficient ones?

D2. Developing Skills of Investigation and Communication

By the end of this course, students will:

- D2.1** use appropriate terminology related to energy conservation and consumption, including, but not limited to: *conventional source, alternative source, efficiency, watt, kilowatt-hour [kWh], joule, BTU, gas meter, electric meter, thermostat, and EnerGuide* [C]
- D2.2** determine the energy consumption of their household over a given time period by reading and interpreting gas and/or electric meters, calculate the cost of consumption (e.g., the number of kWh $\times$ cost per kWh, cubic metres of gas $\times$ cost per cubic metre), and suggest ways in which the household could conserve energy [PR, AI, C]

D2.3 use a research or inquiry process to compare the efficiency of different types or brands of a common household appliance (e.g., different brands of kettles, fans, or refrigerators; natural gas and electric water heaters) or of audio-visual equipment (e.g., different types of computer monitors), and report their findings [IP, PR, AI, C]

D2.4 conduct a risk-benefit analysis of different types of electricity generation (e.g., fossil fuel, hydro, nuclear, wind, and/or solar power) [PR, AI]

D3. Understanding Basic Concepts

By the end of this course, students will:

D3.1 explain the basic principles and characteristics of various types of power generation from non-renewable sources (e.g., coal, oil, natural gas, nuclear) and renewable sources (e.g., hydro-electric, tidal, geothermal, solar, wind, hydrogen fuel cells)

D3.2 compare and contrast renewable and non-renewable energy sources, using criteria such as availability, cost, and environmental impact (e.g., compare a fossil fuel and geothermal energy, using a graphic organizer)

D3.3 describe methods of energy conservation (e.g., the replacement of incandescent bulbs with compact fluorescent bulbs, the replacement of a manual thermostat with a programmable one, the installation of more energy-efficient windows) and some policies that are intended to manage energy demand in the home and the workplace (e.g., variable pricing, which increases the price of electricity during peak hours)

D3.4 describe several criteria used in the construction of energy-efficient buildings (e.g., “smart homes”, in which the use of light, heat, and power for equipment can be programmed; R-2000 homes; straw-bale houses)

E. NATURAL RESOURCE SCIENCE AND MANAGEMENT

OVERALL EXPECTATIONS

By the end of this course, students will:

- E1.** assess the environmental impact of the harvesting and/or extraction of resources, including ways of reducing this impact, and analyse threats to the sustainability of natural resources;
- E2.** investigate methods scientists use to classify and monitor natural resources, and conduct investigations using those methods;
- E3.** demonstrate an understanding of the sustainable use of resources and its relationship to the biodiversity and sustainability of ecosystems.

SPECIFIC EXPECTATIONS

E1. Relating Science to Technology, Society, and the Environment

By the end of this course, students will:

- E1.1** assess the environmental impact of industrial practices related to the extracting or harvesting of natural resources, and describe ways in which that impact can be monitored and minimized [AI, C]

Sample issue: As a result of overfishing, several marine species are endangered. Bottom-trawling drag nets drown sea life, including mammals and turtles, who become entangled in them, and destroy seafloor habitat. In an effort to allow endangered species to recover, governments monitor populations, sometimes limiting catches or declaring moratoriums, and some countries have banned bottom trawling.

Sample questions: What impact can mine tailings have on local water? What practices can be used to reduce this impact? What impact does clear-cutting have on local ecosystems? What impact does large-scale deforestation have on the environment? What harvesting practices can the forestry industry use to minimize the effects of clear-cutting and deforestation?

- E1.2** analyse, on the basis of research, the impact that an environmental contaminant, parasite, or bacteria has on the sustainability of a natural resource in Canada (e.g., the effects of PCBs on Arctic sea mammals, of sea lice on farmed and wild salmon, of *E. coli* on water resources) [IP, PR, AI, C]

Sample issue: As a result of warmer winters and a policy of fire suppression, the mountain pine beetle has decimated coniferous forests in British Columbia, killing millions of lodgepole pines, the most widely harvested tree in the province. There are fears that the beetle will expand into Alberta and could eventually harm pine forests across the country.

Sample questions: How have mercury levels in fish affected the local fishing industry in Northern Ontario? How has mange affected the fox population and people who depend on trapping? What impact has increased bacteria levels in inland waterways had on duck populations?

E2. Developing Skills of Investigation and Communication

By the end of this course, students will:

- E2.1** use appropriate terminology related to natural resources and resource management, including, but not limited to: *population, bioamplification, sampling size, sustainability, ore, mineral, tailings, and succession* [C]
- E2.2** identify and classify a variety of natural resources found in Canada, using appropriate classification systems (e.g., dichotomous keys, botanical keys, tree identification guides, wild-life guides, mineral tests) [PR, AI]

E2.3 investigate, through laboratory inquiry, field study, or simulations, some of the methods and procedures used by scientists to monitor biodiversity in different environments (e.g., making plant tallies in forests; tagging or marking ground vegetation species in fields; tagging and tracking wildlife with the global positioning system in remote areas; using aquatic dip nets for sampling organisms in shallow ponds or streams) [PR]

E2.4 conduct an inventory of a local environment (e.g., a field, a pond), using appropriate techniques and methods (e.g., plant tallies, tags, keys), and display the results graphically [PR, C]

E3. Understanding Basic Concepts

By the end of this course, students will:

E3.1 describe the main types of natural resources found in Canada (e.g., forests, minerals, fisheries, wildlife, water, fossil fuels)

E3.2 describe the characteristics and properties that make a natural resource viable for use (e.g., the size, type, and location of trees; the value,

location, and extraction and processing costs of minerals), and explain the importance of managing natural resources to ensure sustainability and biodiversity

E3.3 describe a variety of methods used to extract or harvest natural resources (e.g., drag nets, strip mining, selective cutting of forests)

E3.4 explain how a variety of sampling techniques (e.g., quadrant sampling, catch-and-release, core sampling to measure tree rings, counting annuli in scales to measure the age of fish) are used to gather information about natural resources

E3.5 explain the importance of biodiversity to the sustainability of life within an ecosystem (e.g., variability among biotic and abiotic factors within an ecosystem decreases the chance that any organism within that ecosystem will become extinct)

E3.6 describe some methods that scientists use to monitor biodiversity in aquatic and terrestrial environments (e.g., field data collection, aerial and satellite imagery)

F. THE SAFE AND ENVIRONMENTALLY RESPONSIBLE WORKPLACE*

OVERALL EXPECTATIONS

By the end of this course, students will:

- F1.** assess workplace situations with respect to safety and environmental issues, and propose a course of action to address unsafe working conditions;
- F2.** investigate a variety of safe and environmentally responsible workplace practices;
- F3.** demonstrate an understanding of general workplace safety procedures and environmentally responsible practices.

SPECIFIC EXPECTATIONS

F1. Relating Science to Technology, Society, and the Environment

By the end of this course, students will:

- F1.1** analyse unsafe working conditions that can affect young workers in various workplace situations (e.g., using information from the Live Safe! Work Smart! website; using information obtained during a co-op placement or through experiential learning), and propose a course of action that would help to improve one such situation [AI, C]

Sample issue: A young worker is asked by her employer to place boxes on the top shelf of an eight-foot storage unit. However, no one is available to help her, and the only ladder in the workplace is unstable. She is unsure what action she should take.

Sample questions: What are the most frequent injuries among young workers? What types of jobs result in the most frequent and/or most serious injuries among young workers? How can the risks associated with these injuries be reduced? How should a young worker address a safety concern? How can companies be encouraged to create safe work environments?

- F1.2** analyse, on the basis of research, and report on the environmental impact of unsafe handling, storage, and disposal of hazardous and non-hazardous workplace materials associated with a particular job [IP, PR, AI, C]

Sample issue: Home construction workers use a range of materials that can harm the environment. Spills of stains and solvents, improper disposal of paint and other chemical substances, the particulate matter created when wall board is cut or insulation is blown, and improper storage of combustible or corrosive materials can contaminate the air, water, and soil.

Sample questions: What impact does the improper storage and disposal of cooking oils in fast-food restaurants have on the environment? In what ways can improper handling or disposal of medical materials (e.g., pharmaceuticals, medical isotopes, disinfectants) in a hospital affect the environment? What is the environmental impact if fast-food restaurants do not separate their waste into compostable, recyclable, and non-recyclable materials?

F2. Developing Skills of Investigation and Communication

By the end of this course, students will:

- F2.1** use appropriate terminology related to safety and environmental responsibility in the workplace, including, but not limited to: *Möbius loop*, *Material Safety Data Sheet (MSDS)*, *Hazardous Household Product Symbols (HHPS)*, *hazardous material*, and *personal protective equipment (PPE)* [C]

* Activities related to the achievement of expectations that refer to “the workplace” and/or that involve hazardous materials may be simulated in the classroom. It is the teacher’s responsibility to ensure students’ safety.

F2.2 demonstrate proper use of a variety of safety techniques and procedures after completing a recognized safety training program (e.g., a “virtual WHMIS” program such as Passport to Safety) [PR]

F2.3 conduct an inventory of hazardous products, safety equipment, and personal protective equipment found in a workplace, using an accepted tool (e.g., a FireSmart Assessment test from the Ministry of Natural Resources), and communicate the results using a table or checklist [PR, C]

F2.4 use appropriate techniques for handling, storing, and disposing of teacher-selected materials, drawing on Material Safety Data Sheets and Canadian Environmental Protection Act regulations (e.g., use appropriate personal protective equipment), and outline proper procedures for handling those materials in the workplace [PR, C]

F2.5 design and report on a plan for reusing, recycling, reducing the volume of, or disposing of a hazardous material found in the workplace (e.g., disposing of batteries, reusing motor or cooking oils for a different purpose) [IP, C]

F2.6 investigate the effectiveness of a personal protective device or environmental protection device for use in the workplace (e.g., compare two different spill kits for absorbing spills; test the key features of a mask for protection from airborne particulate matter; identify the appropriate types of eye protection for different situations) [PR, AI]

(e.g., by reducing the production of garbage and recycling materials for daily use), and explain the meaning of different symbols used to promote these strategies (e.g., different representations of the Möbius loop [the international recycling symbol])

F3.2 compare some of the features, uses, and environmental implications of Hazardous Household Product Symbols and WHMIS hazard symbols

F3.3 identify and describe common types of biological, physical, and chemical hazards in the workplace (e.g., hazards posed by bacteria, noise, work at dangerous heights, use of chemicals and other hazardous materials) and associated accident-prevention methods (e.g., sterilization, soundproofing, use of five-point safety harnesses, use of safe storage cabinets, safe disposal of chemicals)

F3.4 explain how the use of personal protective equipment (e.g., aluminized gloves, a welding shield, ear plugs, a self-contained breathing apparatus, an air-purifying mask) minimizes exposure to hazardous materials that can enter the body through ingestion, inhalation, absorption, and injection

F3.5 identify some current workplace procedures, practices, and protocols that help to protect the environment (e.g., garbage separation, paper recycling, use of recycled products, “telecommuting” to workplaces, practices that conserve water and energy)

F3. Understanding Basic Concepts

By the end of this course, students will:

F3.1 describe some of the ways in which implementation of the 4Rs (reduce, reuse, recycle, and recover) in the workplace protects the environment